

## Fixed reference layer

Future weather  
12 files per pathway

DOE Medium Office  
fixed archetype

Occupancy + people  
real-met heat gain

Same actuator and dwell rules

**Relaxation-first controller  
request setpoint relief  
within actuator bounds**

Guard decision  
relax / hold / protect

## Common evaluation layer

Energy components  
and peak HVAC

PMV / adaptive  
comfort violations

TSV-tail exposure  
zone resolved

**Guard signal is the experimental variable**

Adaptive

PMV

PPD

Mean tail

p90 tail

Exp. TSV

Savings are attributed to threshold-enabled setpoint relaxation, not to changes in the building, weather, or actuator model.